

Bangladesh Telecom Analysis

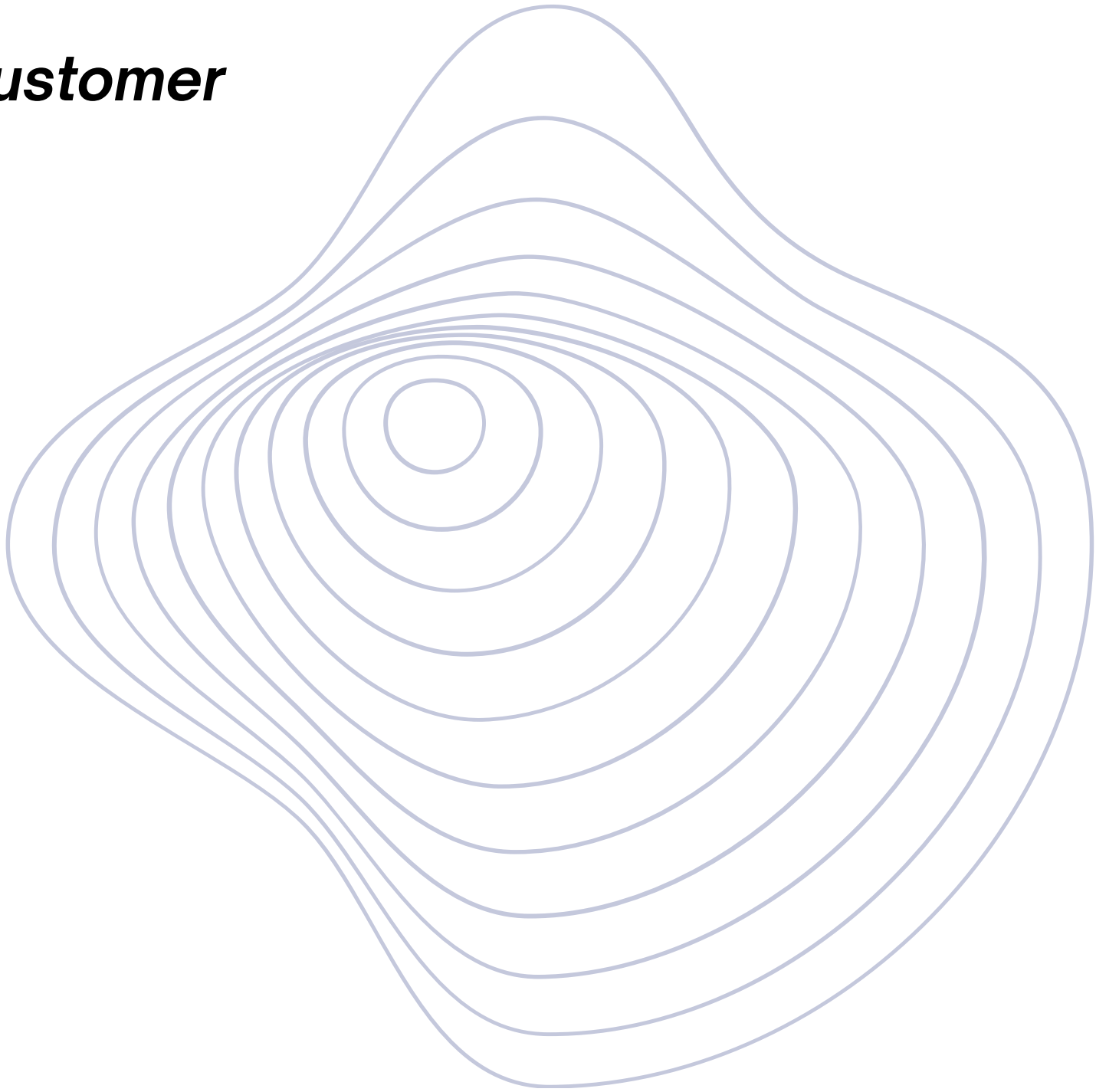
I have performed an end-to-end data analysis on a comprehensive telecom dataset (telecom_customers) containing over 100,000 subscriber records. The goal of this project was to uncover hidden patterns in data usage, regional behavior, customer complaints, and identify the root causes driving customer churn and revenue loss.

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How many customers does each telecom operator have?

```
SELECT  
operator, COUNT(DISTINCT customer_id) AS total_customer  
FROM  
telecom_customers  
GROUP BY operator  
ORDER BY 2 DESC;
```

	operator	total_customer
►	Grameenphone	44978
	Robi	28144
	Banglalink	18047
	Teletalk	4945
	Airtel	3886



Result (Output)

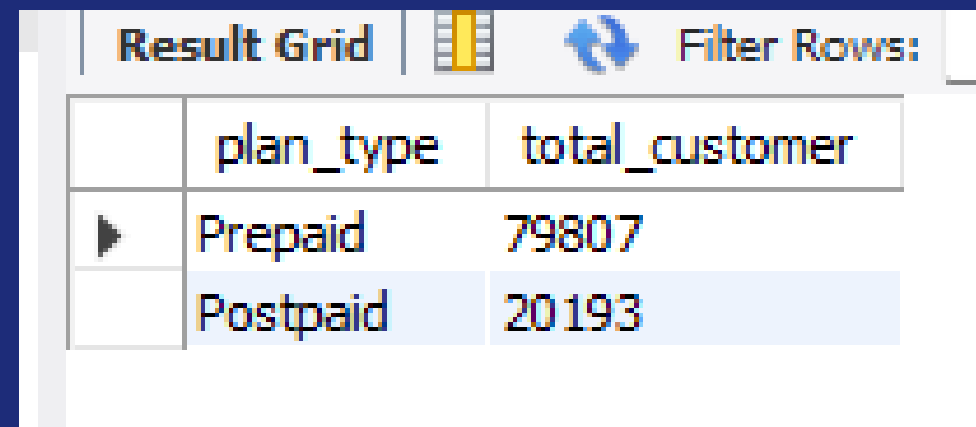
After running the query, it shows that:

Grameenphone has the highest number of customers (44,978).

This is followed respectively by Robi (28,144), Banglalink (18,047), Teletalk (4,945), and Airtel with the lowest (3,886).

Which plan type is the most popular among customers?

```
select plan_type, count(distinct customer_id) as total_customer  
from telecom_customers  
group by 1  
order by 2 desc;
```



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'plan_type' and 'total_customer'. The table has two rows: 'Prepaid' with a value of 79807, and 'Postpaid' with a value of 20193. The 'Prepaid' row is highlighted with a blue background.

	plan_type	total_customer
▶	Prepaid	79807
	Postpaid	20193

Business Action Items

Focus on Prepaid & Launch New Offers: Since the vast majority of customers use Prepaid, attractive recharge offers, internet packs, or minute bundles should be introduced to retain them.

Efforts to Increase Postpaid Customers: The number of Postpaid customers is quite low. To increase this, campaigns can be launched to encourage Prepaid customers to migrate or upgrade to Postpaid by offering attractive benefits (such as zero security deposit, free OTT subscriptions, or family plans).

Customer Retention: Prepaid customers can easily switch operators. Therefore, loyalty programs, reward points, or cashback systems should be introduced to maintain their loyalty.

Which region has the highest number of customers?

```
SELECT
region, COUNT(customer_id) AS highest_number_customer
FROM
telecom_customers
GROUP BY 1
ORDER BY 2 DESC;
```

	region	highest_number_customer
▶	Dhaka	34938
	Chattogram	19875
	Rajshahi	12153
	Khulna	9954
	Sylhet	7990
	Barisal	6088
	Rangpur	4869
	Mymensingh	4133

Business Action Items

- Defend and Optimize Top Markets (Dhaka & Chattogram): Since Dhaka and Chattogram hold the vast majority of your user base, prioritize network capacity upgrades, ensure seamless high-speed coverage, and offer premium localized data/voice bundles here to prevent customer churn.
- Market Penetration in Low-Performing Regions: Regions like Mymensingh, Rangpur, and Barisal show low customer adoption. Launch aggressive, budget-friendly marketing campaigns, lower-priced entry level packages, and expand local retail distribution points to capture market share from competitors.
- Regional Pricing & Campaigns: Tailor promotional offers based on regional economic dynamics. For instance, introduce high-volume entertainment/business packs in high-density areas (Dhaka/Chattogram) and high-validity, cost-effective communication packs in rural or emerging regions.

Which customer segment generates the largest customer base?

```
SELECT  
customer_segment,  
COUNT(customer_id) AS largest_customer_base  
FROM  
telecom_customers  
GROUP BY 1  
ORDER BY 2 DESC;
```

	customer_segment	largest_customer_base
▶	Student	30110
	Business	25090
	Rural	19860
	Urban	14985
	Senior	9955

Business Action Items

Nurture the Student Base (Volume Driver): Since students make up your largest audience, focus on high-volume, low-cost data packs, late-night internet bundles, and digital reward programs (e.g., gaming or music streaming partnerships) to maintain their loyalty.

Maximize Business Segment Value (Revenue Driver): The business segment is substantial and typically offers higher average revenue per user (ARPU). Introduce specialized corporate postpaid plans, dedicated enterprise support, and bulk data/SMS packages tailored for small to large businesses.

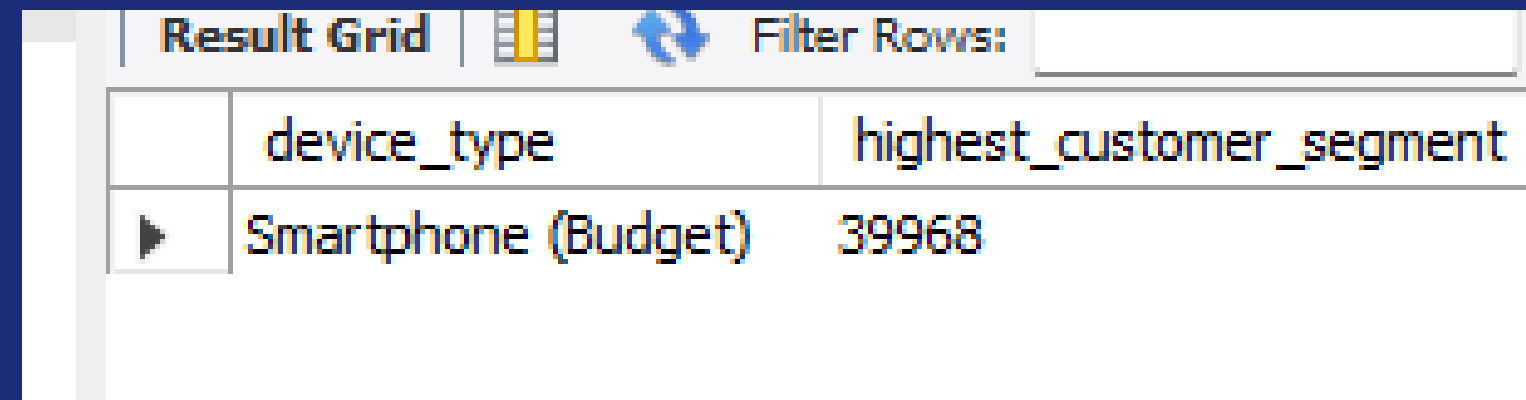
Tailored Strategies for Niche Segments:

For Rural and Urban gaps, analyze the underlying purchasing patterns to push localized campaigns.

For the Senior segment, introduce simplified, voice-heavy, or long-validity packages with easier digital recharge methods to make the service more accessible.

What types of devices are customers using the most?

```
SELECT
    device_type,
    COUNT(customer_id) AS highest_customer_segment
FROM
    telecom_customers
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1
```



The screenshot shows a database query result grid. At the top, there is a tab labeled 'Result Grid' and a 'Filter Rows:' input field. The table has two columns: 'device_type' and 'highest_customer_segment'. The first row shows 'Smartphone (Budget)' with a count of 39968.

	device_type	highest_customer_segment
▶	Smartphone (Budget)	39968

Optimize Digital Platforms for Budget Devices: Ensure that the telecom operator's official mobile app, website, and digital portals are lightweight and fully optimized for lower-end specifications. They should load fast, consume less RAM, and function smoothly on affordable hardware.

Tailor Affordable Digital Products: Since budget smartphone users are typically price-sensitive, focus on promoting cost-effective internet packs, daily/weekly micro-bundles, and cheap social media data tokens (e.g., specific packs for Facebook, YouTube, or TikTok).

Device Bundling Partnerships: Collaborate with popular budget smartphone brands (such as Xiaomi, Realme, or Infinix) to offer co-branded devices bundled with free or discounted data plans. This can attract a high volume of new subscribers looking for maximum value.

What percentage of customers use 3G, 4G, and 5G networks?

```
SELECT
  network_type,
  COUNT(customer_id) as count_customer,
  round((count(customer_id)*100)/sum(count(customer_id)) over()) as percentage
FROM
  telecom_customers
GROUP BY 1
ORDER BY 2 DESC;
```

	network_type	count_customer	percentage
▶	4G	70144	70
	3G	14952	15
	5G	9984	10
	2G	4920	5

Business Action Items

Maintain and Optimize the 4G Backbone: Since 4G accommodates 70% of your total traffic, ensuring high network stability, optimizing speeds during peak hours, and rolling out lucrative 4G data bundles should remain the primary focus to prevent churn.

Accelerate 5G Migration: 5G adoption is still low at 10%. To drive high-value data consumption, target premium 4G customers with 5G-only promotional speed offers, cloud gaming trials, or high-definition streaming bundles to entice them into upgrading.

Phasing Out Legacy Networks (2G & 3G): Together, 2G and 3G capture 20% of your customer base. Because running legacy hardware is expensive, design migration campaigns—such as offering smartphone/SIM upgrade discounts—to shift these users onto 4G networks so you can eventually repurpose that spectrum for better 4G or 5G coverage.



Which internet plan is generating the highest customer adoption?

```
SELECT
  internet_plan,
  COUNT(customer_id) AS highest_customer_adoption
FROM
  telecom_customers
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

	internet_plan	highest_customer_adoption
▶	1GB/day	25077

Business Action Items

- Upsell with Tiered "Daily" Variants: Since the "per day" allocation model is a massive hit, introduce higher-margin daily variants like 1.5GB/day or 2GB/day at a slightly incremental price point to upgrade these existing users and increase average revenue per user (ARPU).
- Introduce Long-Validity Daily Bundles: Lock in these high-adoption customers for longer periods by offering monthly or quarterly subscriptions tied to the same logic (e.g., a 30-day plan offering 1GB/day), perhaps with a small discount compared to buying it daily.
- Targeted Promotional Triggers: Since these users rely on a daily data cap, they are prone to running out of internet during heavy usage. Set up real-time SMS or app notification triggers when they hit 80% of their daily 1GB limit, offering cheap "emergency data add-ons" (like 500MB valid for the rest of the day).



How much monthly revenue is being lost because of churned customers?

```
SELECT
    churned,
    ROUND(SUM(monthly_bill_bdt), 2) AS monthly_lost_revenu
FROM
    telecom_customers
WHERE
    churned = 1
GROUP BY 1;
```

Result Grid			Filter Rows:	
	churned	monthly_lost_revenu		
▶	1	3192101.68		

Top Actions for Recovering Lost Monthly Revenue

- **Fund an Aggressive Win-Back Campaign:** Allocate a dedicated budget to win back recently lost high-value subscribers by offering personalized incentives (e.g., "Reactivate your SIM and get 50% cashback on your first monthly pack").
- **Establish a High-Value Retention Desk:** Create a specialized customer support tier focused exclusively on active users who spend heavily each month, stepping in with tailored offers the moment they exhibit any sign of dissatisfaction or drop in usage.
- **Reinvest in Churn Prevention:** Take a fraction of this 3.19M BDT monthly loss (e.g., 15%) and reinvest it into proactive loyalty rewards, anniversary data gifts, or billing discounts for existing customers to stop them from leaving in the first place.
- **Audit High-Tariff Plan Churn:** Investigate whether this major revenue loss is driven by a small group of premium users churning. If premium users are abandoning the service, adjust the pricing structure or add high-value perks (like complimentary OTT subscriptions) to keep them locked in.



Which payment method is most commonly used by customers?

```
SELECT
    payment_method,
    COUNT(customer_id) AS highest_customer_adoption
FROM
    telecom_customers
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

	payment_method	highest_customer_adoption
▶	Mobile Banking (bKash)	40117

- **Business Action Items**
- **Deepen Integration with bKash:** Since bKash is the primary payment avenue, focus on continuous technical optimizations to minimize payment gateway timeouts and ensure frictionless, one-click recharge capabilities within the telecom app.
- **Launch bKash-Exclusive Cashback/Bonus Offers:** Capitalize on this user preference by launching direct cashbacks, instant data bonuses, or lower-priced internet packs exclusively when purchased via bKash. This reinforces customer retention and usage volume.
- **Co-Marketing Campaigns:** Partner with bKash for joint marketing efforts, such as sending in-app banners or target notifications through the bKash platform to convert inactive SIM cards back into active, paying subscribers.

Which regions generate the highest monthly revenue?

```
SELECT  
    region,  
    ROUND(SUM(monthly_bill_bdt), 2) AS highest_monthly_revenue  
FROM  
    telecom_customers  
GROUP BY 1  
ORDER BY 2 DESC;
```

	region	highest_monthly_revenue
▶	Dhaka	14128241.14
	Chattogram	8062509.63
	Rajshahi	4932488.24
	Khulna	4006082.48
	Sylhet	3268479.43
	Barisal	2467430.73
	Rangpur	1984510.32
	Mymensingh	1661597.09

Business Action Items

Invest Heavily in High-Yield Markets: Since Dhaka and Chattogram bring in the vast majority of cash flow, heavily prioritize capital expenditure (CapEx) in these zones. Double down on corporate enterprise packages, high-speed fiber integrations, and prime 5G network expansion here to maintain high-value billing consistency.

Boost ARPU in Emerging Regions: For mid-tier regions like Rajshahi, Khulna, and Sylhet, design upselling campaigns. Convert lower-paying users into high-value tiers by promoting high-volume entertainment, lifestyle, or business bundles to push their Average Revenue Per User (ARPU) upward.

Cost-Efficient Strategy for Low-Revenue Zones: In Rangpur and Mymensingh, infrastructure should focus on lean operational expenditures (OpEx). Instead of expensive marketing setups, focus on localized, cost-effective digital self-service solutions and basic voice/data combo packs to gradually stimulate transaction volume.

Who are the top 10 customers with the highest monthly bills?

```
SELECT  
customer_id,  
ROUND(SUM(monthly_bill_bdt), 2) AS highest_monthly_revenue  
FROM  
telecom_customers  
GROUP BY 1  
ORDER BY 2 DESC  
LIMIT 10;
```

Result Grid			Filter Rows:
	customer_id	highest_monthly_revenue	
▶	BD0037156	2093.48	
	BD0000033	1920.71	
	BD0045886	1917.45	
	BD0078908	1841.02	
	BD0068707	1824.80	
	BD0028992	1814.59	
	BD0093586	1812.51	
	BD0023404	1788.07	
	BD0085761	1777.88	
	BD0043129	1777.03	

Business Action Items

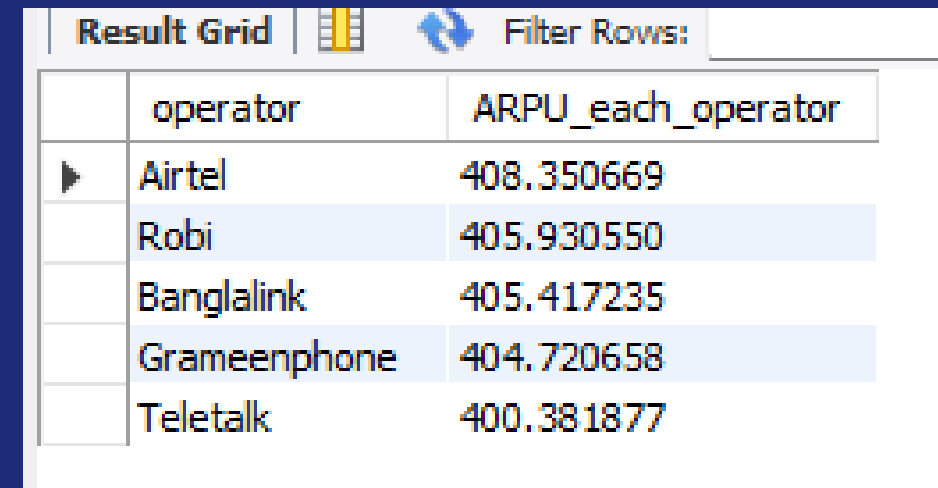
VIP Customer Management (High-Value Tier): These individuals represent your highest Average Revenue Per User (ARPU) segment. Assign them to a premium/VIP support tier with priority call-routing, dedicated relationship managers, or exclusive fast-track service desks to minimize any risk of customer churn.

Exclusive Rewards and Perks: Design a personalized loyalty program tailored specifically for this top spending group. Offering them early access to high-tier network upgrades (like free 5G access trials), lifestyle partner vouchers, or premium data roll-over perks will increase brand stickiness.

Targeted Feedback and Surveys: Directly reach out to this specific customer cluster via personalized surveys to understand what services they use most (e.g., heavy roaming, enterprise tools, or massive data consumption). Use these insights to optimize high-tier bundles for other potential big spenders across the network.

What is the ARPU for each operator?

```
SELECT
    operator,
    (SUM(monthly_bill_bdt) / COUNT(DISTINCT customer_id)) AS ARPU_each_operator
FROM
    telecom_customers
GROUP BY 1 order by 2 desc;
```



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains the following data:

	operator	ARPU_each_operator
▶	Airtel	408.350669
	Robi	405.930550
	Banglalink	405.417235
	Grameenphone	404.720658
	Teletalk	400.381877

Business Action Items

Capitalize on Airtel's High-Value Base: Despite having fewer total subscribers, Airtel has the highest spending average per user. Focus on maintaining this high-value base by rolling out mid-to-high tier value bundles, lifestyle perks, and premium data products to ensure they don't migrate to other networks.

Bridge the ARPU Gap for Market Leaders: For operators like Grameenphone—which traditionally has a massive user volume—even a small shift in ARPU yields enormous returns. Introduce smart upselling techniques, such as minor price increments paired with extra data bonuses, to push their average customer spend past the 405+ mark.

Reposition Teletalk Packages: Since Teletalk is lagging behind at the bottom of the ARPU metric, its existing subscriber packages need a revamp. Transition users away from bare-minimum, low-margin plans toward value-added services (VAS), data combo packs, and digital bundle packages to drive up individual user billing.

Which regions still have the highest SMS usage?

```
SELECT  
region, SUM(sms_count_monthly) AS highest_sms_usage  
FROM  
telecom_customers  
GROUP BY 1  
ORDER BY 2 DESC;
```

	region	highest_sms_usage
▶	Dhaka	1047755
	Chattogram	594550
	Rajshahi	364104
	Khulna	298536
	Sylhet	238916
	Barisal	181820
	Rangpur	146144
	Mymensingh	123781

Strategic Business Actions

Maximize Revenue in High-Traffic Zones (Dhaka & Chattogram):

Launch exclusive, high-volume SMS bundles tailored for heavy users.

Capitalize on B2B opportunities by pitching Bulk SMS marketing services to local businesses, educational institutions, and corporate clients.

Optimize Costs in Low-Traffic Zones (Rangpur & Mymensingh):

Reduce marketing budgets allocated to SMS promotions in these areas.

Pivot to low-cost mobile data and social media bundles, as users here likely prefer internet-based communication (WhatsApp, Messenger).

Shift from paid SMS notifications to free In-App Push Notifications to drastically cut operational costs.



Which internet plans consume the highest amount of data?

```
SELECT
  internet_plan,
  SUM(data_usage_gb_monthly) AS highest_amount_data
FROM
  telecom_customers
GROUP BY 1
ORDER BY 2 DESC limit 1;
```

Result Grid			Filter Rows:	
	internet_plan	highest_amount_data		
▶	1GB/day	74496.490		

Top Actions for "1GB/day" Plan (74.5k+ GB Consumed)

- Upsell to Higher Tiers: Launch and heavily promote 1.5GB/day or 2GB/day plans to these users, as they are likely hitting their daily limits and are ready to pay more for more data.
- Sell "Data Boosters": Offer cheap, instant emergency data add-ons (e.g., 2GB extra for the day) for users who run out of data before midnight.
- Prevent Network Congestion: Optimize network bandwidth and tower capacity in high-usage areas to handle this massive traffic and prevent slow internet speeds.
- Bundle Content: Partner with streaming or gaming apps to offer free subscriptions with the 1GB/day plan, making it more attractive than competitors' offers.

Do customers subscribed to VAS generate higher revenue?

```
SELECT
    vas_subscribed,
    ROUND(SUM(monthly_bill_bdt), 2) AS VAS_generate_higher_revenue
FROM
    telecom_customers
GROUP BY 1;
```

	vas_subscribed	VAS_generate_higher_revenue
	0	28381213.26
	1	12130125.80

Top Actions for VAS Subscription (Value Added Services)

- Launch Free Trials: Convert the massive base of non-VAS users (0) by offering a 7-day or 1-month free trial of popular VAS services (like caller tunes, sports alerts, or gaming portals) to let them experience the value.
- Smart Bundling (Auto-Inclusion): Automatically bundle high-demand VAS features into popular internet packets or voice plans (e.g., "Buy 1GB/day plan, get free sports streaming alerts") to instantly drive up subscription numbers.
- Targeted App Campaigns: Use in-app banners or interactive push notifications specifically targeted at the non-subscribed segment, offering one-click activation discounts.
- Review VAS Catalog: Check if the current VAS offerings are outdated. Retarget the content toward modern trends (like gaming, OTT add-ons, or finance alerts) to attract younger users and boost total revenue.

Does auto-renewal increase customer retention?

```
SELECT
auto_renewal, COUNT(DISTINCT customer_id) AS total_customers
FROM
telecom_customers
GROUP BY 1
ORDER BY 2 DESC;
```

	auto_renewal	total_customers
▶	1	52092
	0	47908

Top Actions for Auto-Renewal Optimization

Incentivize Opt-Ins: Convert manual renewal users (0) by offering an immediate discount or bonus data (e.g., "Turn on auto-renewal and get 10% off your next bill or 2GB free data").

Default Opt-In for New Pack Purchases: Make auto-renewal the default (pre-checked) option whenever a customer purchases a new pack or plan via the app or website, allowing them to manually opt-out if they wish.

Low-Balance Alerts with Auto-Renew Prompts: Send a push notification or SMS right when a manual user's balance or package is running low, offering a seamless one-click switch to turn on auto-renew.

Exclusive Tier Rewards: Introduce a loyalty tier or "Set & Forget" badges for auto-renewing users that unlocks exclusive monthly scratch cards or priority customer service to prevent churn.



Which customers have not recharged for the longest time?

```
SELECT  
customer_id, MAX(last_recharge_date) AS longest_time  
FROM  
telecom_customers  
GROUP BY 1  
ORDER BY 2 DESC  
LIMIT 1;
```

	customer_id	longest_time
▶	BD00002161	2024-12-31

What are the most common complaint types reported by customers?

```
SELECT
  complaint_type,
  COUNT(DISTINCT customer_id) AS most_complaint_types
FROM
  telecom_customers
GROUP BY 1
ORDER BY 2 DESC;
```

Result Grid			Filter Rows:
	complaint_type	most_complaint_types	
▶	No Complaint	65233	
	Network Issue	9892	
	Billing Error	7953	
	Customer Service	7058	
	Data Speed	5878	
	Call Drop	3986	

Top Actions for Complaint Resolution & Quality Improvement

- Fix Core Network Infrastructure: Prioritize engineering resources to resolve infrastructure bottlenecks, as "Network Issue" is the number one active complaint. Target specific regions showing high failure rates to reduce this number.
- Audit and Fix Billing Systems: Since "Billing Error" is the second largest issue, conduct an immediate audit of the automated charging systems, hidden fees, or incorrect package renewals to stop unintended charges.
- Train and Automate Customer Support: Optimize "Customer Service" workflows by introducing AI chatbots for instant ticket resolution and retraining support agents to handle billing and network queries faster.
- Address Data Speed & Call Drops: Combine "Data Speed" and "Call Drop" optimizations by upgrading tower bandwidths in heavy traffic areas to tackle the remaining technical service complaints.

Which regions generate the highest number of complaints?

```
SELECT
region, COUNT(complaint_type) AS highest_complaint_region
FROM
telecom_customers
GROUP BY 1
ORDER BY 2 DESC;
```

	region	highest_complaint_region
▶	Dhaka	34938
	Chattogram	19875
	Rajshahi	12153
	Khulna	9954
	Sylhet	7990
	Barisal	6088
	Rangpur	4869
	Mymensingh	4133

Top Actions for Regional Complaint Mitigation

- Deploy Ground Teams to Dhaka & Chattogram:** Allocate the majority of network engineers and field technicians to these two cities to address the root causes of physical network bottlenecks and infrastructure stress.
- Scale Up Customer Support Capacity:** Increase dedicated customer care agents and localized support desk capacity specifically for Dhaka and Chattogram callers to reduce wait times and improve resolution rates.
- Proactive Compensation Campaigns:** Run automated apology campaigns offering small data tokens or bill discounts to highly impacted users in Dhaka and Chattogram to prevent high-volume customer churn.
- Monitor Mid-Tier Vulnerabilities:** Audit Rajshahi (~12k) and Khulna (~10k) systems to ensure their growing complaint counts are managed before they scale to critical levels.

Which customer segments have the highest and lowest NPS scores?

```
SELECT
nps_score, COUNT(customer_id) AS highest_lowest_nps
FROM
telecom_customers
GROUP BY 1 order by 2 desc;
```

	nps_score	highest_lowest_nps
	4	10190
	3	10152
	6	10092
	9	10004
	2	9998
	10	9975
	5	9969
	1	9969
	7	9948
	8	9703

Top Actions for NPS Improvement & Retention

- Launch a Detractor Recovery Program: Prioritize reaching out to the highly concentrated customer segments scoring between 1 and 6 (especially scores 3 and 4) through automated surveys or support calls to uncover and fix their specific pain points.
- Convert Passives into Promoters: Target the users scoring 7 and 8 (nearly 20k customers combined) with exclusive loyalty rewards, early product access, or surprise perks to easily push them into the Promoter category (9-10).
- Leverage Promoters for Advocacy: Incentivize the satisfied segments scoring 9 and 10 (~20k customers) with a referral program (e.g., "Refer a friend and get free data"), turning them into a low-cost organic marketing channel.
- Investigate Systemic Dissatisfaction: Since the scores are spread out so uniformly, it suggests a widespread mix of experiences. Correlate these specific NPS segments with data from previous network complaints or billing errors to find out why so many users are rating the service poorly.

Identify customers with low tenure, low NPS, and high complaints who are likely to churn.

```
SELECT
tenure_months, nps_score, churned, customer_id
FROM
telecom_customers
WHERE
churned = 1 AND tenure_months <= 5
AND nps_score <= 4
ORDER BY nps_score DESC;
```

	tenure_months	nps_score	churned	customer_id
▶	5	4	1	BD0000018
	5	4	1	BD0000097
	1	4	1	BD0000169
	5	4	1	BD0000657
	1	4	1	BD0000691
	2	4	1	BD0000886
	4	4	1	BD0077361
	4	4	1	BD0096646
	1	4	1	BD0001333
	4	4	1	BD0001399
	5	4	1	BD0076662
	1	4	1	BD0001634
	3	4	1	BD0001908
	4	4	1	BD0002033
	2	4	1	BD0002091
	4	4	1	BD0002165
	1	4	1	BD0074354

What is the overall churn rate of the telecom company?

```
select churned,count(customer_id) as Total_customers,  
       (count(customer_id)*100)/sum(count(customer_id))  
       over() as churn_rate from telecom_customers  
group by 1 order by 2 desc;
```

	churned	Total_customers	churn_rate
▶	0	91732	91.7320
	1	8268	8.2680

Top Actions for Churn Reduction & Customer Retention

Deploy Win-Back Campaigns: Target the 8,268 churned users (1) with highly aggressive, automated win-back offers via SMS or direct calls (e.g., "Come back and get a 50% discount on your next internet plan").

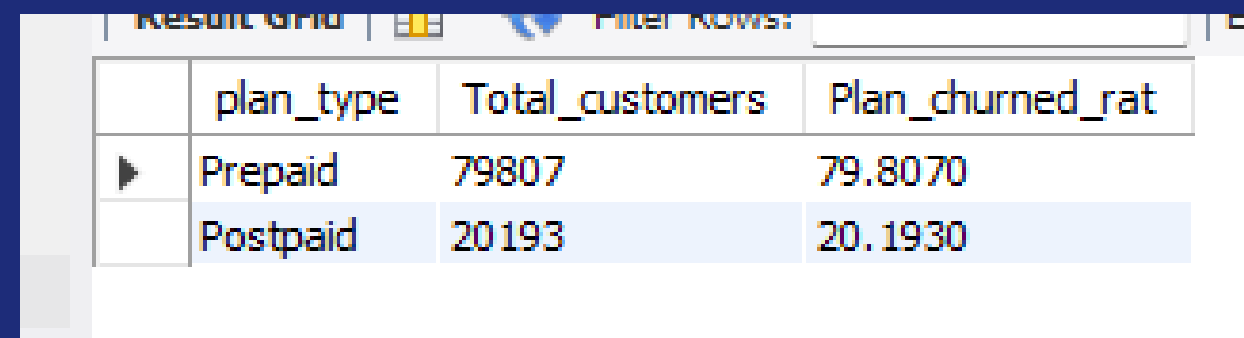
Set Up a Predictive Churn Model: Use the data profiles of the existing churned users to build an early-warning system that flags active customers showing similar behavioral patterns (like a sudden drop in data usage) before they actually leave.

Exit Surveys for Root Cause Analysis: Conduct quick, targeted automated exit surveys for users who have recently stopped renewing packages to identify whether price, network issues, or competitor offers caused them to churn.

Double Down on Loyalty Programs: Strengthen rewards for the 91.73% active user base (0) by introducing automated milestone rewards (e.g., anniversary bonus data) to keep them highly engaged and insulated from competitor switching tactics.

Which plan types have the highest churn rates?

```
select plan_type, count(customer_id) as Total_customers,  
(count(customer_id)*100)/sum(count(customer_id))over() as  
Plan_churned_rat from telecom_customers  
group by 1;
```



	plan_type	Total_customers	Plan_churned_rat
▶	Prepaid	79807	79.8070
	Postpaid	20193	20.1930

Top Actions for Plan-Type Churn Mitigation

- **Fix the Prepaid Leakage:** Since Prepaid users represent nearly 80% of the churn, introduce aggressive re-engagement campaigns like instant recharge bonuses or heavily discounted "welcome back" packs to stop them from dropping off.
- **Incentivize Prepaid-to-Postpaid Migration:** Run targeted campaigns offering low-cost, hassle-free migration to entry-level Postpaid plans for high-value Prepaid users to increase long-term customer lock-in.
- **Optimize Postpaid Contract Retention:** Though Postpaid churn is lower (~20%), these are high-revenue users. Review their billing experiences and proactively offer contract renewal perks (e.g., free family sharing data add-ons) before their current cycles expire.
- **Gamify Prepaid Recharges:** Introduce localized scratch cards or in-app spin-the-wheel rewards for every successful Prepaid recharge to keep the highly volatile Prepaid subscriber base consistently engaged.

Which telecom operator is losing the most customers?

```
SELECT
operator, COUNT(churned) AS Total_lossing_churned
FROM
telecom_customers
WHERE
churned = 1
GROUP BY 1
ORDER BY 2 DESC;
```

Result Grid			Filter Rows:
	operator	Total_lossing_churned	
▶	Grameenphone	3723	
	Robi	2315	
	Banglalink	1503	
	Teletalk	414	
	Airtel	313	

Top Actions for Operator-Specific Churn Mitigation

- Targeted Retention for Grameenphone Users: Since Grameenphone has the highest churn volume (3,723), immediately analyze their primary pain points—such as pricing or network performance—and deploy aggressive loyalty offers specifically for this segment.
- Competitive Poaching Campaigns for Competitors: If you are a competing operator (like Banglalink or Teletalk), launch aggressive marketing campaigns directly targeting dissatisfied Grameenphone and Robi users with attractive Mobile Number Portability (MNP) switch-over benefits.
- Price & Value Audit for Robi Segment: Review the data and voice package pricing for the Robi segment (2,315 churned) to determine if competitor price drops are driving these users away, and adjust mid-tier package values accordingly.
- Maintain Safety Margins for Lower Churn Tiers: While Teletalk (414) and Airtel (313) have significantly lower churn numbers, closely monitor their user retention metrics to ensure their churn patterns do not start scaling upward.